

Tanya Bhat

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RESEARCH INTERESTS

Curious and motivated Cognitive Science major at the University of California, San Diego, specializing in Machine Learning and Neural Computation with a minor in Computer Science. Experienced in Python, ML frameworks, Deep Learning and Multi-agent LLM systems. Eager to contribute to advancements in human–AI interaction and accessible technology.

EDUCATION

University of California, San Diego (UCSD)

B.S. in Cognitive Science Spec. Machine Learning and Neural Computation | Expected: June 2028

Minor in Computer Science

GPA: 3.93

Relevant Coursework: Object Oriented Programming in Java, Mathematics for Algorithms, Calculus I & II, Statistics and Research Methods, Japanese (ACTFL Intermediate Low-Mid Proficiency)

TECHNICAL SKILLS

Languages: Python, Java

Libraries/Tools: PyTorch, NumPy, Pandas, Matplotlib, Pygame, Kivy, Pymunk

PROJECTS

AI Song Recommender | *Team Project* | June 2025 - September 2025

- Collaborated in a team of four to build a song recommendation system using LLM-based semantic text embeddings.
- Applied neural network and vector representation concepts to match written input with musically relevant tracks.

Multi-Agent LLM Therapy System | *Research Project, UCSD* | September 2025 - Present

- Developing a therapy-inspired system where multiple LLMs collaborate to analyze conversations.
- Using LangFlow and LangChain for agent orchestration and model interaction.
- Implementing in Python with multiple LLM APIs to explore applications of multi-agent AI in mental health contexts.

EMG Piano | *UCSD Neurotech Club* | September 2025 - Present

- Using EMG electrodes to control a virtual piano for accessibility applications
- Working in a 10-person team to collect EMG data and train models using deep learning and reinforcement learning

Research Paper: NPCs in Game Design | *Lumiere Research Scholars' Program* | June 2023 - October 2023

 **Social_Presence_NPCs.docx.pdf**

- Wrote a systematic review paper exploring how NPCs can be used to increase player social presence. Combined insights from AI, psychology, and game design; received A+ for the work

Happiiminds – Prototype for Educational App for Autistic Children | *Team project* | August 2023 - January 2024

- Created interactive games and animations to gamify social skill learning using Kivy (Python)
- Researched user needs, tested in therapy centers
- Worked in a team of four and presented at inter-school prototype challenge (4th place)

Malaria Detection with CNNs | *Veritas AI Scholars Program* | January 2023 - February 2023

- Gained conceptual understanding of ML and data preprocessing using NumPy, Matplotlib, etc.
- Built a basic Convolutional Neural Network to classify malaria-infected cells with 92% accuracy

Brella – 2D Platformer Game | High school project | January 2023 - May 2024

- Designed and built a game with custom storyline, composed music, 2D animations and robot battle mechanics
- Integrated physics elements like projectiles, collisions, and gravity using Pygame and Pymunk
- Received 'Best CS Project award' for my work